Retail Customer Behavior Analysis

PROJECT OVERVIEW

This project analyzes retail customer shopping behavior from a 3.9K-instance financial and transactional dataset, ranging from different product categories (e.g., Clothing, Footwear, Outerwear, Accessories, etc.). The goal is to gain insights into customers' spending patterns, product preferences, segments, and subscription behavior to guide strategic business decisions.

DATASET SUMMARY

- Shape: 3900 Rows x 18 Columns
- Key Features:
  - Customer demographics (Age, Gender, Location, Subscription Status)
  - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
  - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data:
  - 37 values in the "Review Rating" column

EXPLORATORY DATA ANALYSIS

Data preparation and cleaning were done in Python:

- **Data Loading:** Imported the data using `pandas`
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics
- **Missing Data:** Checked for null values and filled in missing values in the `Review Rating` column using the median for each respective product category
- **Column Standardization:** Standardized column names to snake case (i.e., *Item Purchased* became *item_purchased*)
- **Feature Engineering:**
  - Created `age_group` column by categorizing customer ages
  - Created `purchase_frequency_days` column from purchase data

- **Data Consistency Check:** Verified that `discount_applied` and `promo_code_used` were redundant--dropped `promo_code_used` as a result
- **Database Integration:** Connected Python script to PostgreSQL server, loading the cleaned, processed DataFrame into the SQL database for analysis in the next section

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SQL DATA ANALYSIS (BUSINESS QUESTIONS)

1.) What is the total revenue generated by male vs. female customers?

```
SELECT
     gender,
     SUM(purchase_amount) AS revenue
FROM customer_purchases
GROUP BY gender;
```

	gender text	revenue numeric
1	Female	75191
2	Male	157890

2.) Which customers used a discount but still spent more than the average purchase amount?

```
SELECT customer_id, purchase_amount
FROM customer_purchases
WHERE discount_applied = 'Yes' AND purchase_amount >=
     (SELECT AVG(purchase_amount)
      FROM customer_purchases);
```

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97

3.) Which are the top 5 products with the highest average review rating?

```
SELECT
     item_purchased,
     ROUND(AVG(review_rating::numeric), 2) AS "Average
     Product Rating"
FROM customer_purchases
GROUP BY item_purchased
ORDER BY AVG(review_rating) DESC
LIMIT 5;
```

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4.) Compare the average purchase amounts between standard and express shipping.

```
SELECT
     shipping_type,
     ROUND(AVG(purchase_amount), 2)
FROM customer_purchases
WHERE shipping_type in ('Standard', 'Express')
GROUP BY shipping_type;
```

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

5.) Do subscribed customers spend more? Compare average spend and total revenue between subscribers and non-subscribers.

```
SELECT
     subscription_status,
     COUNT(customer_id) AS total_customers,
     ROUND(AVG(purchase_amount), 2) AS avg_spend,
     ROUND(SUM(purchase_amount), 2) AS total_revenue
FROM customer_purchases
GROUP BY subscription_status
ORDER BY total_revenue, avg_spend DESC;
```

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

6.) Which 5 products have the highest percentage of purchases with discounts applied?

```
SELECT
     item_purchased,
     ROUND(100 * SUM(CASE WHEN discount_applied = 'Yes'
     THEN 1 ELSE 0 END) / COUNT(*), 2) AS discount_rate
FROM customer_purchases
GROUP BY item_purchased
ORDER BY discount_rate DESC
LIMIT 5;
```

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

7.) Segment customers into New, Returning, and Loyal based on their total number of previous purchases, and show the count of each segment.

```
WITH customer_type AS (
     SELECT
          customer_id,
          previous_purchases,
          CASE
               WHEN previous_purchases = 1 THEN 'New'
               WHEN previous_purchases BETWEEN 2 AND 10
               THEN 'Returning'
               ELSE 'Loyal'
               END AS customer_segment
          FROM customer_purchases
)

SELECT
     customer_segment,
     COUNT(*) AS 'Num of customers'
FROM customer_type
GROUP BY customer_segment;
```

	customer_segment text	Num of customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

8.) What are the top 3 most purchased products within each category?

```
WITH item_counts AS (
     SELECT
          category,
          item_purchased,
          COUNT(customer_id) AS total_orders,
          ROW_NUMBER() OVER (
               PARTITION BY category
               ORDER BY COUNT(customer_id) DESC
          ) AS item_rank
     FROM customer_purchases
```

```
    GROUP BY category, item_purchased
)

SELECT item_rank, category, item_purchased, total_orders
FROM item_counts
WHERE item_rank <= 3;
```

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171

9.) Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?

```
SELECT
    subscription_status,
    COUNT(customer_id) AS repeat_buyers
FROM customer_purchases
WHERE previous_purchases > 5
GROUP BY subscription_status;
```

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

10.) What is the revenue contribution of each age group?

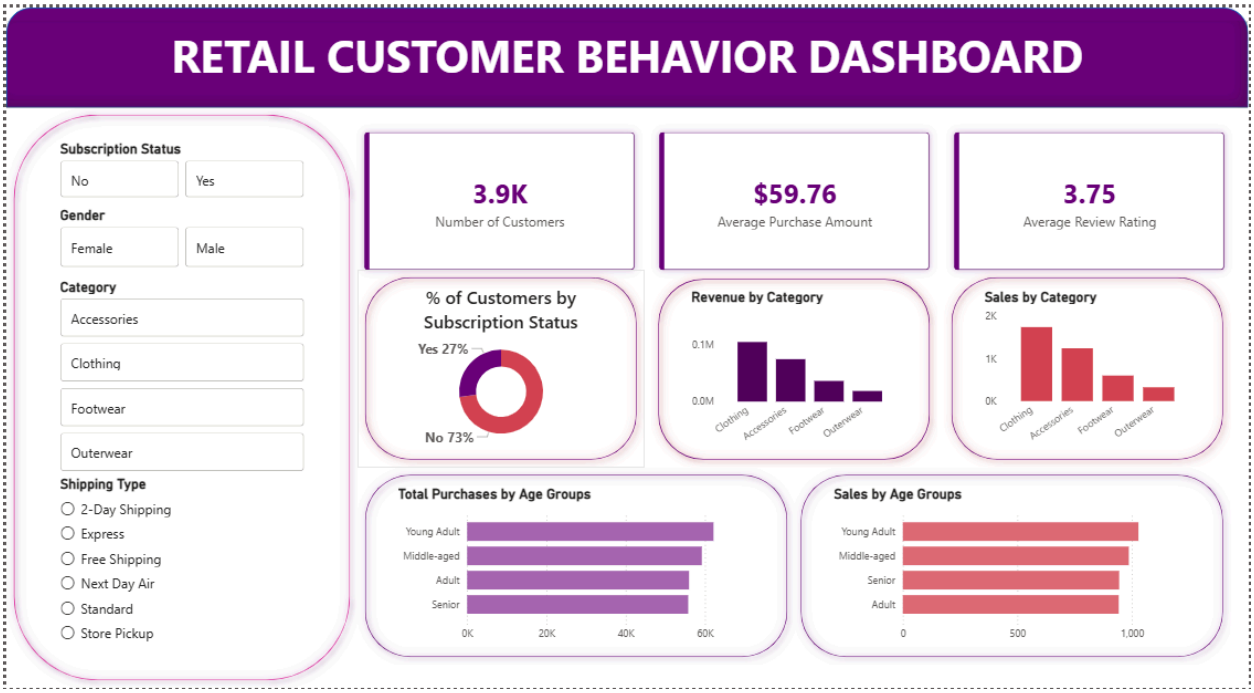
```
SELECT
    age_group,
    SUM(purchase_amount) AS total_revenue,
FROM customer_purchases
```

```
GROUP BY age_group
ORDER BY total_revenue DESC;
```

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

POWER BI DASHBOARD

An interactive Power BI dashboard was built to visually present our insights.



FINAL BUSINESS RECOMMENDATIONS

- ➢ Promote exclusive benefits for subscribers
- ➢ Reward repeat buyers to move them into the “Loyal” customer segment
- ➢ Balance sales boosts with margin control; review the discount policy
- ➢ Highlight top-rated and best-selling products in campaigns
- ➢ Focus efforts on high-revenue age groups (young adult and middle-aged) and express-shipping users